```
#include <iostream>
using namespace std;
#define size 5
int arr[size];
int top=-1;
void push(int val){
   if (top>=4)
   {
    cout<<"stack overflow"<<endl;
   }
   else {
    top++;
    arr[top]=val;
    cout<<val<<"pushed"<<endl;
   }
}
void pop()
{
    if (top==-1){
        cout<<"stack is empty"<<endl;
    }
    else{
        cout<<arr[top]<<"popped"<<endl;
    }

}
void peak()
{
    if (top==-1){
        cout<<"stack is empty"<<endl;
    }
    else{
        cout<<arr[top]<<"this is top most element"<<endl;
    }

}
void display()
```

```
{
    if (top==-1){
        cout<<"empty"<<endl;
    }
    else{
        for (int i=top;i>=0;i--){
            cout<<arr[i];
            cout<<" ";
        }

    }
    cout<<endl;
}
int main ()
{
    push(10);
    push (20);
    push(30);
    display();
    pop();
    peak();
    push(40);
    pop();
    pop();
    peak();
    push(10);
    display();
    return 0;
}
```